

Distributed solar advantages in a nutshell

No transmission losses. Grid-connected multi-megawatt solar farms at 11kV and above not only suffer from usual distribution loss of 15 per cent (20 per cent on the power grid, not counting 10 per cent additional loss in theft) but also further loss of three to four per cent in stepping up the expensive solar power generated at 400V.

Distributed solar, on the other hand, can use existing low-voltage transmission in a local substation, avoiding this 20 to 25 per cent loss of high-cost solar power. Distributed solar power of up to 2-300 kW is produced at a low voltage (400V AC, three-phase) and can be either distributed by a local mini-grid (if away from the National Power Grid) or fed into the substation of the National Power Grid.

Less expensive. There are no economies of scale in solar power generation from multi-megawatt solar plants. Costs of PV solar plants scale linearly from kilowatts to megawatts. So project management and land cost make multi-megawatt solar plants more expensive.

No restraint on location. Solar photovoltaic (PV) technology is clean and quiet, which means it can be sited on or within a locality. Distributed solar is also easy to adapt to site conditions and scalable to existing transmission capacity.

Mini grid protocol. Many small plants can be progressively linked via the small generator interconnection protocol (SGIP). The result is that distributed-scale projects go through a quicker review and are less likely to encounter huge delays and bureaucratic wrangles waiting for lengthy power grid extension.

Lower cost than extending National Power Grid. The 2003 plan of the Ministry of Power in India to extend the grid power was abandoned because the investment required per house was Rs 23,000. In 2012, it could be even double. With local mini-grid, this cost will be much below Rs 20,000. More importantly, the energy is free of cost.

Grid stability. A large part of distributed solar power is consumed locally. Multi-megawatt plants, on the other hand, supply 100 per cent to the grid. Managing the grid becomes difficult with increase in solar penetration.

Less use of environmentally-sensitive land. Unlike large-scale solar projects, distributed-scale projects don't need thousands of acres of pristine wilderness. Instead, distributed-scale projects can be sited on large industrial rooftops, unused terraces and backyards. This means they get through land use approvals and permissions faster.

High-voltage grid feed of solar farm power has no benefits. Energy from 2-50MW PV solar plants fed into the existing power grid is just a tiny fraction of power flowing through the grid. So it in no way improves the supply. This small addition makes little difference in terms of solving the problems of blackouts and brownouts that consumers suffer. Hours of power loss will continue.

Low-voltage grid feed of distributed power helps 100 per cent use of solar power. PV solar is relatively expensive and when it is fed into the grid at EHT level, it suffers high transmission loss of over 25 per cent. So the cost of useful solar energy goes up by 25 per cent. In distributed solar plants, almost all the power generated is available for use.

Distributed solar helps to create small power SMEs. Distributed solar power from 50kW-300kW solar plants creates opportunities for SME investment in solar power generation. This will trigger local employment and open avenues for new local entrepreneurs.

land resource adding to ecological problems, and also the cost?

Solar energy incentives in most developed solar markets in the US and Europe have clearly shifted the preference to distributed, small, rooftop solar installations on residences. This, because they reduce the need for expensive power-generation infrastructure, improve reliability and put money in the hands of the common citizens. The US, Spain, Germany and Italy—the four biggest markets in the world—have done this.

India, however, has so far not focussed on rooftop solar installations. This has to change since this approach is both, a great environmental strategy and a sound business strategy. While our individual solar projects would be smaller than large-scale projects, one has to remember that mini power plants today deliver over 1800 MW across North America and Europe.

In short, for the National Solar Mission 2020 to succeed, policy makers will have to view solar power

generation from a developmental perspective. Energy security and rural electrification will get a huge shot in the arm from solar power projects that can quickly come up across the country as part of the current policy drive. Efforts are also required to promote community-based awareness, especially in rural areas.

Present situation

Solar energy is perfectly suited for India compared to many other sunlight-starved countries. The government has made a start by fixing an ambitious target of 22GW solar capacity by 2022, which rivals that of China. However, the first phase of the Jawaharlal Nehru National Solar Mission (JNNSM) has mainly given subsidies and incentives to multi-megawatt solar installations and inexplicably ignored more apt, distributed solar, widely promoted in countries like the US, the UK and Germany. As a result, in the last two years, India saw a flurry of multi-megawatt (2-200MW) PV solar power plant announcements—mostly with foreign technology and designs. Despite this, the JNNSM is nowhere near its 2012 target.

India desperately needs more renewable energy. The need is urgent for hundreds of millions of homes and businesses in places where the power grid has not reached.

According to government reports, 400 million Indians today have no access to grid power. Giving them just a solar lantern will not serve any purpose. There is no better way to give electrical power to them than to exploit the local availability of sun, wind and biomass, by focusing on distributed small-scale projects. Distributed-scale projects enjoy important advantages over large-scale projects. These are less complex, faster to interconnect and bypass time-consuming land-use issues that often add years to project timelines. ●

The author was formerly chairman of Electronics Commission of the government of India, and electronics advisor to India's late prime minister Rajiv Gandhi